Domain Adaptation Based on Correlation Subspace Dynamic Distribution Alignment for Remote Sensing Image Scene Classification

Jun Zhang, Jiao Liu, Bin Pan, *Member, IEEE*, and Zhenwei Shi, *Member, IEEE*

***Abstract*— Remote sensing image scene classification refers to assigning semantic labels according to the content of the remote sensing scenes. Most machine learning-based scene classification methods assume that training and testing data share the same distributions. However, in real application scenarios, this assumption is difficult to guarantee. Domain adaptation (DA) is a promising approach to address this problem by aligning the feature distribution of training and testing data. Inspired by the idea DA, in this article, we propose a correlation subspace dynamic distribution alignment (CS-DDA) method for remote sensing image scene classification. Aiming at the characteristics of remote sensing scenes, we introduce two strategies to balance the effects of source and target domains: subspace correlation maximization (SCM) and dynamic statistical distribution alignment (DSDA). On the one hand, SCM tries to avoid mapping source domain data into irrelevant subspace to preserve the representation information of the source domain. On the other hand, DSDA is proposed to reduce the data distribution discrepancy between aligned source and target domains. Specifically, DSDA is a dynamic adjustment process where an adaptive factor is learned to balance the interclass and intraclass distribution between domains. Moreover, we integrate SCM and DSDA into a uniform optimization framework, and the optimal solution can be converted to the generalized eigendecomposition problem by derivation. The experimental results indicate that the proposed method can generate better results when compared with other feature distribution alignment methods.**

***Index Terms*— Data shift, distribution alignment, domain adaptation (DA), remote sensing image scene classification.**

I. Introduction

With the development of satellite sensing technology, the task of remote sensing image interpretation has drawn significant attention, such as scene classification, hyperspectral classification, and semantic segmentation [1]–[4]. As a basic image understanding work, remote sensing image scene classification aims to infer semantic labels according to the content of remote sensing scenes [5], [6], which is beneficial to resource management, urban planning, and environment monitoring [7]–[9].

During the past decades, extensive efforts have been made to develop remote sensing image scene classification methods. Early works on remote sensing scene classification are mainly based on handcrafted feature representations, such as scale-invariant feature transform [10] and bag-of-visual-words model [11]. Nevertheless, due to highly complex geometrical structures and spatial patterns in remote sensing scene images, handcrafted features may fail to capture high-level semantic information of remote sensing scene images [12], [13]. To address this problem, convolutional neural networks (CNNs) were used for scene classification of remote sensing images. References [14]–[17] trained new CNN models from scratch using remote sensing images. Cheng *et al.* [1], Castelluccio *et al.* [18], and Scott *et al.* [19] fine-tuned the pretrained CNNs on the remote sensing images. Fang *et al.* [20], Weng *et al.* [21], Han *et al.* [22], and Dai *et al.* [23] exploit the pretrained CNNs as a feature extractor to extract the high-level features for remote sensing images. Especially, features can be extracted from any layer of a pretrained network; references [24]–[29] fuse the feature of different layers.

The aforementioned methods usually assumed that the training and testing data shared the same distribution. However, in a real application, due to the influence of sensors, geographic locations, imaging conditions, and other factors, the distribution of training and testing data may be different. This phenomenon is referred to as the data shift. When there is a data shift between the training set and the test set, classification models have to be reconstructed from scratch using the newly collected training data [30]. Obviously, it is expensive to collect annotation data and rebuild the model [31]. To weaken the influence of data shift, domain adaptation (DA) algorithms were proposed. DA is one of the transfer learning approaches, which tries to remove the data shift between source and target domains [32], [33]. In DA, we define one data set with plenty of labeled samples as the source domain, and another data set, which is collected under different imaging conditions, as the target domain.

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Jun Zhang and Jiao Liu are with the School of Artificial Intelligence, Hebei University of Technology, Tianjin 300401, China, and also with the Hebei Province Key Laboratory of Big Data Calculation, Hebei University of Technology, Tianjin 300401, China (e-mail: zhangjun@scse.hebut.edu.cn; liujiao.hebut@hotmail.com).

Bin Pan is with the School of Statistics and Data Science, Nankai University, Tianjin 300071, China (e-mail: panbin@nankai.edu.cn).

Zhenwei Shi is with Image Processing Center, School of Astronautics, Beihang University, Beijing 100191, China (e-mail: shizhenwei@buaa.edu.cn).

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In the remote sensing scene classification, researchers have proposed several methods to alleviate data shifts based on DA. Othman *et al.* [34] presented a DA network to deal with the data shift problem in remote sensing image scene classification. Reference [35] introduced an asymmetric adaptation neural network method for cross-domain classification. Teng *et al.* [36] proposed a classifier constrained deep adversarial DA method for cross-domain semisupervised classification. In [37], subspace alignment (SA) is designed and a CNN-based framework was used to solve the data shift problem. The abovementioned methods tried to reduce the distribution difference by aligning the global distribution of the two domains; however, they did not consider the distribution alignment between categories. Mathematically, the global distribution is called marginal distribution, and the class distribution is called conditional distribution.

Zhang *et al.* [38] proposed the joint geometrical and statistical alignment (JGSA)-based method to reduce the data shift. Different from the methods that only considering marginal distribution, JGSA learned two projections that transform the source domain and target domain data into respective subspaces where the marginal distribution, conditional distribution, and geometrical divergence are aligned simultaneously. Inspired by the idea of multidistribution ensemble, in this article, we develop a new DA method under the framework of JGSA.

However, JGSA is originally designed for the task of natural scene object recognition, which is quite different from scene classification. In remote sensing image scene classification task, JGSA has at least two shortcomings that may hinder its application:

1) Remote sensing scene images present strong interclass similarity. Besides removing the diversity between source and target domains, it is quite necessary to enhance the discrepancy of corresponding categories.
2) The weights between marginal and conditional distributions in JGSA are fixed. Although JGSA considers both of the distributions, it cannot evaluate the importance of them.

In this article, to overcome these limitations, we propose a correlation subspace dynamic distribution alignment (CS-DDA) method for remote sensing image scene classification. Two strategies are developed to improve JGSA: subspace correlation maximization (SCM) and dynamic statistical distribution alignment (DSDA). SCM attempts to prevent mapping the source domain data into irrelevant subspace, which preserves the information structure in the source domain. Meanwhile, we propose DSDA to balance the influence of marginal and conditional distributions. DSDA exploits an adaptive factor to adjust class distribution alignment. In particular, this balanced factor can be estimated according to the data distribution between the source domain and the target domain.

The main contributions of this article can be summarized as follows.

1) A new DA method is proposed for remote sensing image scene classification, where CS-DDA is developed.
2) We propose SCM to avoid mapping the source domain data into unrelated subspaces.
3) We design DSDA that aims to eliminate the influence of distribution weights via learning an adaptive factor.

The rest of this article is organized as follows. Section II introduces the overall framework of the proposed method as well as our two contributions. The description of data sets, experimental setup, experimental results, and feature distribution analysis are presented in Section III. Section IV concludes this article.

II. Methodology

This section describes CS-DDA for remote sensing scene classification, which contains algorithm background, SCM, DSDA, and final objective function. The process of remote sensing scene classification is shown in Fig. 1.

A. Algorithm Background

JGSA [38] is a representative algorithm for DA, so we first introduce the work of JGSA.

The labeled source domain data are denoted by $\mathcal{D}_s = \{x_i, y_i\}_{i=1}^{n_s}$, where $x_i \in \mathbb{R}^d$ is the deep feature of the source domain image I_i, n_s is the number of source domain samples, $y_i \in \{1, 2, \ldots, C\}$ is the corresponding category label, and C is the number of categories. The source domain matrix $X_s = \{x_i\}_{i=1}^{n_s}$ is drawn from the distribution $P_s(X_s)$. The unlabeled target domain data can be defined as follows: $\mathcal{D}_t = \{x_j\}_{j=1}^{n_t}$, and its matrix $X_t = \{x_j\}_{j=1}^{n_t}$ is drawn from the distribution $P_t(X_t)$. The feature spaces and label spaces between domains are the same: $\mathcal{X}_s = \mathcal{X}_t$ and $\mathcal{Y}_s = \mathcal{Y}_t$. Due to the data shifts, $P_s(X_s) \neq P_t(X_t)$. The previous method assumes a unified transformation $P_s(\phi(X_s)) = P_t(\phi(X_t))$, $P_s(Y_s|\phi(X_s)) = P_t(Y_t|\phi(X_t))$. Nevertheless, when the divergence of the two domains is large, such a unified transformation may not exist. Therefore, JGSA finds two projections M and N to generate a new feature representation of the source and target domains.

To prevent mapping features of the target domain into irrelevant dimensions, the variance of the target domain is maximized

$$\max_N \operatorname{Tr}\left(N^T S_t N\right) \tag{1}$$

where $S_t = X_t H_t {X_t}^T$ is the scattering matrix of the target domain, $H_t = I_t - (1/n_t)1_t 1_t^T$ is the centering matrix, and $1_t \in \mathbb{R}^{n_t}$ is the column vector with all elements one.

Especially, JGSA exploits the label information of the source domain to make the features discriminative

$$\max_M \operatorname{Tr}\left(M^T S_b M\right) \tag{2}$$

$$\min_M \operatorname{Tr}\left(M^T S_w M\right) \tag{3}$$

where S_b is the interclass scattering matrix of the source domain and S_w is the intraclass scattering matrix of the source

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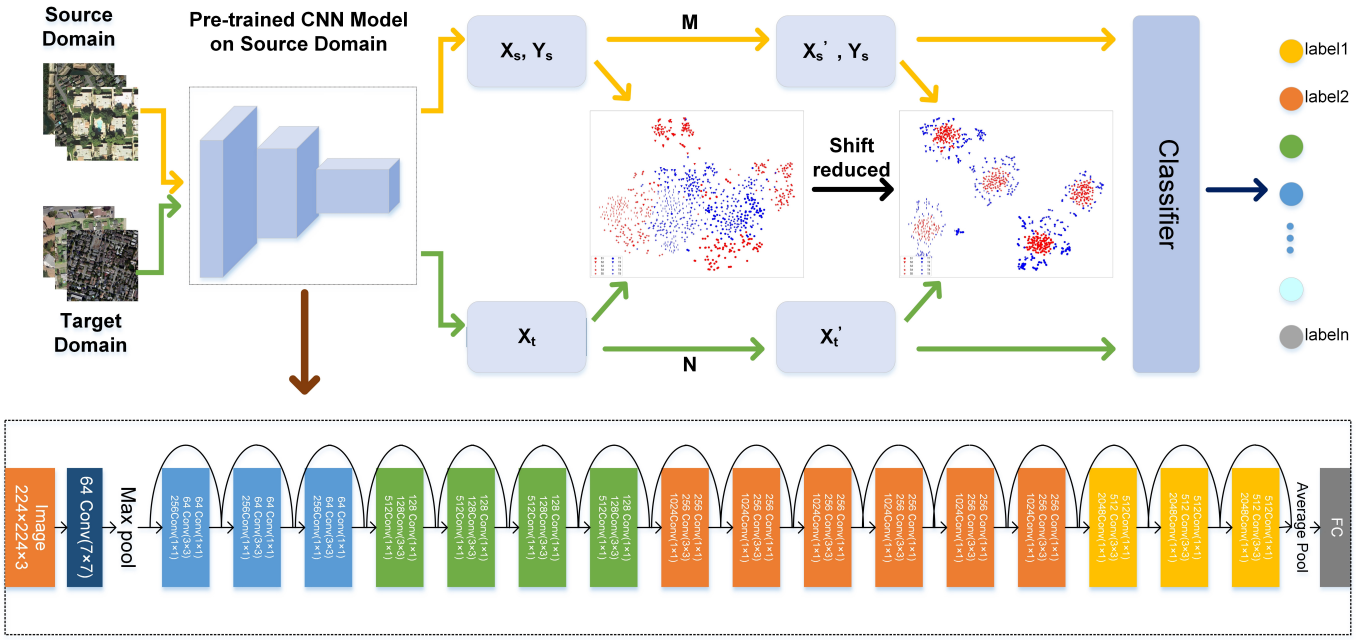


Fig. 1. Overall architecture of the proposed method. First, a pretrained CNN on source domain is applied to extract features of the source domain and target domain. Then, to alleviate the data shift, we leverage CS-DDA to handle the original features X_s and X_t. Furthermore, a classifier is trained by using the new feature representation X_s' in the source domain. Finally, the trained classifier f is utilized to classify X_t'.

domain, which is defined as follows:

$$S_w = \sum_{c=1}^{C} X_s{}^{(c)} H_s^{(c)} \left(X_s{}^{(c)}\right)^T \tag{4}$$

$$S_b = \sum_{c=1}^{C} n_s^{(c)} \left(a_s^{(c)} - \overline{a}_s\right)\left(a_s^{(c)} - \overline{a}_s\right)^T \tag{5}$$

where $X_s{}^{(c)} \in \mathbb{R}^{d \times n_s^{(c)}}$ is the set of source samples belonging to class c, $a_s^{(c)} = 1/n_s^{(c)} \sum_{i=1}^{n_s^{(c)}} x_i^{(c)}$, $\overline{a}_s = 1/n_s \sum_{i=1}^{n_s} x_i$, $H_s^{(c)} = I_s^{(c)} - (1/n_s^{(c)}) 1_s^{(c)} \left(1_s^{(c)}\right)^T$ is the centering matrix of data within class c, $I_s^{(c)} \in \mathbb{R}^{n_s^{(c)} \times n_s^{(c)}}$ is the identity matrix, $1_s \in \mathbb{R}^{n_s}$ is the column vector with all elements one, and $n_s^{(c)}$ is the number of source samples in class c.

To reduce the difference between marginal distributions of source and target domains, JGSA adopts maximum mean discrepancy (MMD) [39] as the distance measure to compare different distributions, which measures the distance between the sample means of the source and target data in the subspace

$$\min_{M,N} \left\| \frac{1}{n_s} \sum_{\mathbf{x}_i \in X_s} M^T \mathbf{x}_i - \frac{1}{n_t} \sum_{\mathbf{x}_j \in X_t} N^T \mathbf{x}_j \right\|_F^2. \tag{6}$$

However, reducing the discrepancy in the marginal distributions cannot guarantee that the conditional distributions are also be removed [40]. Hence, JGSA leverages the pseudo labels of the target domain to minimize the conditional distribution differences between the source domain and the target domain

$$\min_{M,N} \sum_{c=1}^{C} \left\| \frac{1}{n_s^{(c)}} \sum_{\mathbf{x}_i \in X_s{}^{(c)}} M^T \mathbf{x}_i - \frac{1}{n_t^{(c)}} \sum_{\mathbf{x}_j \in X_t{}^{(c)}} N^T \mathbf{x}_j \right\|_F^2. \tag{7}$$

Moreover, JGSA removes the geometric discrepancy between domains by making the subspaces of the source domain and the target domain close

$$\min_{M,N} \|M - N\|_F^2. \tag{8}$$

B. SCM

JGSA utilizes the label information to constrain the new representation of source domain data. Nevertheless, for the remote sensing image scene, source domain data may be mapped into unrelated subspaces. The reason is that the category of remote sensing image scene is related to not only the category of objects but also the background and spatial layout.

To avoid projecting source domain data into unrelated subspaces, we maximize the source domain variance. The variance maximization can be calculated as follows:

$$\max_{M} \operatorname{Tr}\left(M^T S_s M\right) \tag{9}$$

where

$$S_s = X_s H_s X_s{}^T \tag{10}$$

is the scattering matrix of the source domain. In (10), H_s is the central matrix, which is defined as follows:

$$H_s = I_s - \frac{1}{n_s} 1_s 1_s^T \tag{11}$$

where $I_s \in \mathbb{R}^{n_s \times n_s}$ is the identity matrix, and $1_s \in \mathbb{R}^{n_s}$ is the column vector with all elements one.

C. DSDA

Different from natural object images, remote sensing scene images often exhibit complex spatial structures with intraclass diversity and interclass similarity. Therefore, for remote sensing scene images, it is not appropriate to alleviate the marginal

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distribution and the conditional distribution discrepancy with equal weight as JGSA.

In this article, we propose a DSDA algorithm to adaptively adjust the importance of the conditional distribution according to the feature distribution of the remote sensing scene. We add the factor α to control the conditional distribution. It can be represented as follows:

$$\min_{M,N}\left\|\frac{1}{n_s}\sum_{\mathbf{x}_i\in X_s}M^T\mathbf{x}_i-\frac{1}{n_t}\sum_{\mathbf{x}_j\in X_t}N^T\mathbf{x}_j\right\|_F^2 +\alpha\sum_{c=1}^{C}\left\|\frac{1}{n_s^{(c)}}\sum_{\mathbf{x}_i\in X_s{}^{(c)}}M^T\mathbf{x}_i-\frac{1}{n_t^{(c)}}\sum_{\mathbf{x}_j\in X_t{}^{(c)}}N^T\mathbf{x}_j\right\|_F^2 \tag{12}$$

where the first term denotes the marginal distribution distance between domains and the second term is the conditional distribution distance. According to the relationship between the matrix norm and the matrix trace, (12) can be rewritten as follows:

$$\min_{M,N}\mathrm{Tr}\left(\begin{bmatrix}M^T & N^T\end{bmatrix}\begin{bmatrix}R_s & R_{st}\\ R_{ts} & R_t\end{bmatrix}\begin{bmatrix}M\\ N\end{bmatrix}\right) \tag{13}$$

where

$$R_s = X_s\left(Q_s+\alpha\sum_{c=1}^{C}Q_s^{(c)}\right)X_s{}^T,\ Q_s=\frac{1}{n_s^2}1_s1_s^T$$

$$\left(Q_s^{(c)}\right)_{ij}=\begin{cases}\dfrac{1}{\left(n_s^{(c)}\right)^2}, & \mathbf{x}_i,\mathbf{x}_j\in X_s{}^{(c)}\\ 0, & \text{otherwise}\end{cases} \tag{14}$$

$$R_t = X_t\left(Q_t+\alpha\sum_{c=1}^{C}Q_t^{(c)}\right)X_t{}^T,\ Q_t=\frac{1}{n_t^2}1_t1_t^T$$

$$\left(Q_t^{(c)}\right)_{ij}=\begin{cases}\dfrac{1}{\left(n_t^{(c)}\right)^2} & \mathbf{x}_i,\mathbf{x}_j\in X_t{}^{(c)}\\ 0 & \text{otherwise}\end{cases} \tag{15}$$

$$R_{st} = X_s\left(Q_{st}+\alpha\sum_{c=1}^{C}Q_{st}^{(c)}\right)X_t{}^T,\ Q_{st}=-\frac{1}{n_sn_t}1_s1_t^T$$

$$\left(Q_{st}^{(c)}\right)_{ij}=\begin{cases}-\dfrac{1}{n_s^{(c)}n_t^{(c)}}, & \mathbf{x}_i\in X_s{}^{(c)},\mathbf{x}_j\in X_t{}^{(c)}\\ 0, & \text{otherwise}\end{cases} \tag{16}$$

$$R_{ts} = X_t\left(Q_{ts}+\alpha\sum_{c=1}^{C}Q_{ts}^{(c)}\right)X_s{}^T,\ Q_{ts}=-\frac{1}{n_tn_s}1_t1_s^T$$

$$\left(Q_{ts}^{(c)}\right)_{ij}=\begin{cases}-\dfrac{1}{n_s^{(c)}n_t^{(c)}}, & \mathbf{x}_j\in X_s{}^{(c)},\mathbf{x}_i\in X_t{}^{(c)}\\ 0, & \text{otherwise}\ .\end{cases} \tag{17}$$

However, there is a distinct disadvantage in (12): the factor α has to be manually fixed, which cannot balance the influence of marginal and conditional distributions in remote sensing scenes. Aiming at this problem, in CS-DDA, we propose an adaptive factor adjustment approach based on $\mathcal{A}$-distance [41], [42]. The $\mathcal{A}$-distance is defined as follows:

$$d_A(\mathcal{D}_s,\mathcal{D}_t)=2(1-2\epsilon(h)) \tag{18}$$

where h is the classifier and $\epsilon(h)$ denotes the error of classifier discriminating the source domain and the target domain.

Thus, we estimate α based on the $\mathcal{A}$-distance. In this process, we utilize the global and local information of the domain, which can be calculated as follows:

$$\hat{\alpha}\approx\frac{\sum_{c=1}^{C}d_c\left(\mathcal{D}_s^{(c)},\mathcal{D}_t^{(c)}\right)}{d_m(\mathcal{D}_s,\mathcal{D}_t)} \tag{19}$$

where d_c denotes the conditional $\mathcal{A}$-distance, d_m denotes the marginal $\mathcal{A}$-distance, $\mathcal{D}_s^{(c)}$denotes samples from class c in $\mathcal{D}_s$, and $\mathcal{D}_t^{(c)}$ denotes samples from class c in $\mathcal{D}_t$.

D. Final Objective Function

In remote sensing image scene classification, we require to generate an effective and robust feature representation. Hence, we incorporate (1)–(3), (8), (9), (12), and (19) to determine the final objective function. The final objective function can be formalized as follows:

$$\max_{M,N}\frac{\beta(9)+\gamma(1)+\delta(2)}{(12)+\lambda(8)+\delta(3)}. \tag{20}$$

In (20), minimizing the denominator has two meanings: one is to reduce the statistical and geometric differences between the source domain and the target domain and the other is to minimize the intraclass variance of the source domain. In addition, maximizing the numerator of (20) encourages the large target domain variance, source domain variance, and the interclass variance of the source domain. Among them, $\beta,\gamma,\delta,\lambda$ are factors that balance the importance of each part.

Furthermore, we follow [43] to constrain N. To optimize (20), we write $\begin{bmatrix}M^T & N^T\end{bmatrix}$ as Z^T, so the objective function can be written as follows:

$$\max_{Z}\frac{\mathrm{Tr}\left(Z^T\begin{bmatrix}\beta S_s+\delta S_b & 0\\ 0 & \gamma S_t\end{bmatrix}Z\right)}{\mathrm{Tr}\left(Z^T\begin{bmatrix}R_s+\lambda I+\delta S_w & R_{st}-\lambda I\\ R_{ts}-\lambda I & R_t+(\lambda+\delta)I\end{bmatrix}Z\right)}. \tag{21}$$

We discover that scaling Z does not affect the objective function, so (21) can be rewritten as follows:

$$\max_{Z}\mathrm{Tr}\left(Z^T\begin{bmatrix}\beta S_s+\delta S_b & 0\\ 0 & \gamma S_t\end{bmatrix}Z\right)$$
$$\text{s.t }\mathrm{Tr}\left(Z^T\begin{bmatrix}R_s+\lambda I+\delta S_w & R_{st}-\lambda I\\ R_{ts}-\lambda I & R_t+(\lambda+\delta)I\end{bmatrix}Z\right)=1. \tag{22}$$

Actually, (22) is a constrained optimization problem, and it is convenient to apply Lagrange techniques. Hence, the corresponding Lagrangian function for (22) is given as follows:

$$\mathcal{L}=\mathrm{Tr}\left(Z^T\begin{bmatrix}\beta S_s+\delta S_b & 0\\ 0 & \gamma S_t\end{bmatrix}Z\right) +\mathrm{Tr}\left(\left(Z^T\begin{bmatrix}R_s+\lambda I+\delta S_w & R_{st}-\lambda I\\ R_{ts}-\lambda I & R_t+(\lambda+\delta)I\end{bmatrix}Z-I\right)\Theta\right). \tag{23}$$

By setting the derivative of $\partial\mathcal{L}/\partial Z=0$, we can convert it into an eigendecomposition problem

$$\begin{bmatrix}\beta S_s+\delta S_b & \mathbf{0}\\ \mathbf{0} & \gamma S_t\end{bmatrix}Z=\begin{bmatrix}R_s+\lambda I+\delta S_w & R_{st}-\lambda I\\ R_{ts}-\lambda I & R_t+(\lambda+\delta)I\end{bmatrix}Z\Theta \tag{24}$$

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where Θ is the eigenvalues matrix and Z is the corresponding eigenvectors matrix. Therefore, we can acquire the map matrices M and N by Z. Finally, the source domain data and the target domain data are mapped into the new subspaces to get the new feature representation $X_s{}' = M^T X_s$ and $X_t{}' = N^T X_t$. The pseudocode of CS-DDA is shown in Algorithm 1.

Algorithm 1 CS-DDA Algorithm

Input:

Data: Source domain matrix X_s, target domain matrix X_t, source domain label Y_s

Parameter: $\gamma = 1, \delta = 1, \lambda = 1, \beta = 1, \alpha$

Output:

Projection matrix: M, N

New feature representation: $X_s{}'$, $X_t{}'$

Target domain label Y_t

Train a classifier using original X_s, and apply prediction on X_t to get the initial pseudo-label $Y_{t'}$

Calculate

$S_t = X_t H_t {X_t}^T$, $S_s = X_s H_s {X_s}^T$

$S_w = \sum_{c=1}^{C} {X_s}^{(c)} H_s^{(c)} \left({X_s}^{(c)}\right)^T$

$S_b = \sum_{c=1}^{C} n_s^{(c)} \left(a_s^{(c)} - \overline{a}_s\right)\left(a_s^{(c)} - \overline{a}_s\right)^T$

repeat

Estimate α by $\hat{\alpha} \approx \frac{\sum_{c=1}^{C} d_c\left(\mathcal{D}_s^{(c)}, \mathcal{D}_t^{(c)}\right)}{d_m(\mathcal{D}_s, \mathcal{D}_t)}$

Compute

$R_s = X_s\left(Q_s + \alpha \sum_{c=1}^{C} Q_s^{(c)}\right) {X_s}^T$

$R_t = X_t\left(Q_t + \alpha \sum_{c=1}^{C} Q_t^{(c)}\right) {X_t}^T$

$R_{st} = X_s\left(Q_{st} + \alpha \sum_{c=1}^{C} Q_{st}^{(c)}\right) {X_t}^T$

$R_{ts} = X_t\left(Q_{ts} + \alpha \sum_{c=1}^{C} Q_{ts}^{(c)}\right) {X_s}^T$

Solve the generalized eigendecomposition problem in Eq. (24), and obtain projection matrices M, N by Z

Construct the new feature representation $X_s{}' = M^T X_s$ and $X_t{}' = N^T X_t$

Train a new classifier on $X_s{}'$ to update the pseudo-label of $Y_{t'}$

Update α, R_s, R_t, R_{st}, R_{ts}

until Convergence

return Target domain label Y_t

III. Experiment

In this section, we introduce the data set description, experimental setup, experimental results, and feature distribution analysis.

A. Data Sets

To verify our algorithm, we select the UC Merced, AID, NWPU-RESISC45, and RSSCN7 to build the cross-domain remote sensing image scene data sets.

1) *UC Merced Data Set [11]:* The UC Merced data set is a widely used data set for remote sensing image scene classification and can be downloaded from the United States Geological Survey National Map of 20 U.S. regions. It consists of 2100 remote sensing images from 21 scene classes: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium-density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. Each scene class contains 100 red–green–blue (RGB) images with a pixel resolution of 1 ft and an image size of 256 × 256 pixels.
2) *AID Data Set [44]:* The AID data set is a large-scale aerial image data set and acquired from Google Earth. It contains 10 000 images with a size of 600 × 600 pixels, which are divided into 30 classes, including airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaducts. The number of each class varies from 220 to 420, and the pixel resolution changes from 8 m to half a meter.
3) *NWPU-RESISC45 Data Set [1]:* The NWPU-RESISC45 data set consists of 31 500 remote sensing images that are divided into 45 scene classes. These 45 scene classes are as follows: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Each class includes 700 images with a size of 256 × 256 pixels in the RGB color space. The spatial resolution varies from about 30 to 0.2 m per pixel for most of the scene classes.
4) *RSSCN7 Data Set [45]:* The RSSCN7 data set contains 2800 remote sensing scene images, which are from seven typical scene categories, namely, the grassland, forest, farmland, parking lot, residential region, industrial region, and river and lake. There are 400 images in each scene type, and each image has a size of 400 × 400 pixels. It is worth noticing that the sample images in each class are sampled on four different scales with 100 images per scale with different imaging angles.

From the above description of the data set, we discover that these four data sets have five public categories, namely, farmland, forests, dense residential areas, rivers, and parking lot. Hence, we choose these five categories to do the cross-domain remote sensing image scene data sets. Some example images are shown in Fig. 2.

B. Experimental Setup

In this article, we establish three cross-domain scenarios termed as UCM→RSSCN7, AID→RSSCN7,

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Fig. 2. Sample images from four public data sets used in our experiments (five classes in total). (Top row to the bottom row) UC Merced data set, the AID data set, the NWPU-RESISC45 data set, and the RSSCN7 data set. Each column presents the corresponding class of these four data sets. (From left to right) Dense residential, farmland, forest, parking lot, and river.

and NWPU→RSSCN7 referring to source domain→target domain. To analyze the generalization ability of CS-DDA on the three cross-domain remote sensing scene data sets, two classic pretrained CNN models (i.e., AlexNet [46] and ResNet50 [47]) are utilized to extract the features of the source and target domains. Specifically, the features of both source domain data and target domain data before the final fully connected (FC) layer are extracted [37]. The dimensions of features extracted from AlexNet and ResNet50 are 4096 and 2048, respectively. For the parameter setting of CS-DDA, we follow JGSA.

To evaluate the proposed algorithm, we select overall accuracy (OA), class accuracy (CA), kappa coefficient, and confusion matrix as evaluation metrics.

In addition, to verify the CS-DDA, we compare it to the following.

1) *1-Nearest Neighbor (NN):* A basic baseline learns NN classifier on the source domain and applies to the target, which could be viewed as a lower bound.
2) *Transfer Component Analysis (TCA) [32]:* TCA tries to learn some transfer components across domains to reduce the marginal distribution difference.
3) *SA [48]:* SA aligns the base of the source domain with the base of the target domain by finding a transformation.
4) *CORrelation ALignment (CORAL) [49]:* CORAL minimizes domain shift by aligning the second-order statistics of source and target distributions.
5) *Joint Distribution Analysis (JDA) [50]:* JDA attempts to jointly adopt both the marginal distribution and conditional distribution.
6) *Transfer Joint Matching (TJM) [51]:* TJM aims to reduce the domain discrepancy by jointly matching the features and reweighting the instances.
7) *JGSA:* A baseline of CS-DDA.

C. Experimental Results

In this section, we comprehensively evaluate CS-DDA and seven baseline methods on three cross-domain scenarios. All experimental results are illustrated in Table I, where the best results are marked in bold.

1) Results on NWPU→RSSCN7: Our first experiment was conducted on NWPU→RSSCN7, including two cases. In the first case of AlexNet as a feature extractor, CS-DDA achieves much better performance than all other baselines, with the OA of 84.40. Especially, the OA and Kappa of CS-DDA improve by 2.25 and 2.81, respectively, compared with the best baseline JGSA. There is a reasonable explanation of why CS-DDA performs better than JGSA. CS-DDA searches for relevant subspace as much as possible and adjusts the conditional distribution according to the data distribution of the source domain and the target domain. To analyze the generalization ability of CS-DDA on the NWPU→RSSCN7, we also adopt ResNet50 as a feature extractor. Similar to the first case, CS-DDA achieves favorable results in the case of ResNet50 as a feature extractor.

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TABLE I
CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THREE CROSS-DOMAIN DATA SETS (%)

Data set	Feature	Class	Methods							
			NN	TCA	SA	CORAL	JDA	TJM	JGSA	CS-DDA
NWPU→RSSCN7	AlexNet	Dense residential	74.25	82.25	80.25	77.00	81.75	82.25	**89.50**	83.50
		Farmland	49.50	69.50	72.00	73.50	72.25	81.75	81.75	**87.50**
		Forest	**96.25**	84.25	86.50	88.50	90.75	87.00	83.25	83.25
		Parking lot	68.25	78.25	81.00	**84.25**	79.75	76.25	71.25	77.50
		River	32.25	72.50	72.00	66.00	80.75	79.00	85.00	**90.25**
		Kappa	55.13	71.69	72.94	72.31	76.44	76.56	77.69	**80.50**
		OA	64.10	77.35	78.35	77.85	81.15	81.25	82.15	**84.40**
	ResNet50	Dense residential	79.75	**86.25**	79.50	78.00	84.50	83.50	82.25	81.00
		Farmland	73.00	84.25	83.75	86.50	85.25	85.25	**88.75**	88.25
		Forest	**93.25**	79.75	89.50	87.75	83.25	85.50	91.00	93.00
		Parking lot	92.50	89.75	95.00	94.75	89.00	86.50	**95.75**	95.00
		River	47.75	60.75	61.75	73.75	72.25	67.50	81.25	**85.75**
		Kappa	71.56	75.19	77.38	80.19	78.56	77.06	84.75	**85.75**
		OA	77.25	80.15	81.9	84.15	82.85	81.65	87.80	**88.60**
AID→RSSCN7	AlexNet	Dense residential	89.00	88.25	85.50	87.75	79.75	84.75	**90.75**	82.25
		Farmland	81.75	74.25	70.00	78.75	83.25	76.75	**91.00**	90.25
		Forest	77.50	77.75	80.75	77.25	83.75	84.25	87.00	**88.25**
		Parking lot	30.25	47.50	53.50	46.00	**76.75**	67.25	58.75	74.00
		River	51.00	77.75	80.75	84.75	86.75	86.5	90.50	**91.50**
		Kappa	57.38	66.38	67.63	68.63	77.56	74.88	79.50	**81.56**
		OA	65.90	73.10	74.10	74.90	82.05	79.90	83.60	**85.25**
	ResNet50	Dense residential	78.75	82.00	83.00	**89.00**	80.25	83.75	78.75	79.00
		Farmland	53.50	83.25	81.50	**92.00**	89.00	87.50	83.50	87.00
		Forest	**94.75**	83.00	73.50	86.00	77.25	77.50	89.50	89.50
		Parking lot	68.00	67.75	71.50	64.00	77.75	74.25	**93.25**	93.00
		River	52.00	74.50	72.75	78.25	82.25	82.75	82.75	**84.00**
		Kappa	61.75	72.63	70.56	77.31	76.63	76.44	81.94	**83.13**
		OA	69.40	78.10	76.45	81.85	81.30	81.15	85.55	**86.50**
UCM→RSSCN7	AlexNet	Dense residential	28.50	48.50	47.00	59.50	68.75	61.50	74.75	**80.50**
		Farmland	58.50	54.50	73.75	39.75	79.00	79.25	85.50	**87.75**
		Forest	**95.50**	84.75	83.75	63.75	89.50	88.00	84.50	83.00
		Parking lot	27.50	49.00	47.00	34.25	65.25	**66.25**	52.75	53.25
		River	48.75	89.25	80.50	**95.50**	84.50	79.00	92.75	91.50
		Kappa	39.69	56.50	58.00	48.19	71.75	68.50	72.56	**74.00**
		OA	51.75	65.20	66.40	58.55	77.40	74.80	78.05	**79.20**
	ResNet50	Dense residential	49.75	56.25	49.75	68.50	83.25	69.25	**84.75**	**84.75**
		Farmland	61.75	69.00	74.00	58.00	**85.00**	81.75	77.75	78.25
		Forest	90.50	84.75	74.00	71.00	87.00	90.25	**91.50**	91.25
		Parking lot	50.00	69.75	53.50	48.25	81.00	69.00	79.50	**79.75**
		River	72.75	81.75	86.75	88.75	74.75	77.75	85.50	**85.75**
		Kappa	56.19	65.38	59.50	58.63	77.75	72.00	79.75	**79.94**
		OA	64.85	73.20	67.60	66.90	82.20	77.60	83.80	**83.95**

It is worth noting that almost all feature distribution alignment methods are better than the methods of NN. This phenomenon indicates the importance of mitigating the data shift when the source domain and target domain are drawn from different distributions. The OA and Kappa of the JDA are better than TCA. The reason is that JDA simultaneously adapts both marginal and conditional distributions. CS-DDA and JGSA are better than JDA because they preserve the discrimination information of the source domain. In particular, TJM is better than TCA because TJM combines marginal distribution matching and reweighting the instances. In addition, the OA and Kappa of SA have not shown good performance. The reason is that SA fails to minimize the distributions between domains after aligning the subspaces.

In the first case of AlexNet as a feature extractor, for "Farmland" and "River," CS-DDA acquires a better CA compared with JGSA. In Fig. 3, we provide the confusion matrix for both cases in NWPU→RSSCN7. From Fig. 3, we find that the performance of the ResNet50 is better than that of AlexNet. However, some categories are confused, such

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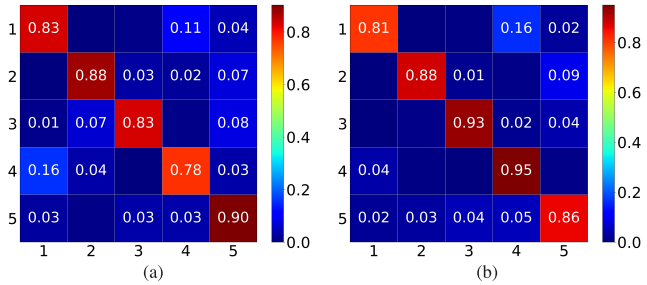


Fig. 3. Confusion matrix on the NWPU→RSSCN7. (a) AlexNet as a feature extractor. (b) ResNet50 as a feature extractor. The rows and columns of the matrix denote the actual and predicted classes, respectively. The corresponding category labels of 1–5 are dense residential, farmland, forest, parking lot, and river. The color bar indicates the proportion of samples over the actual total class samples.

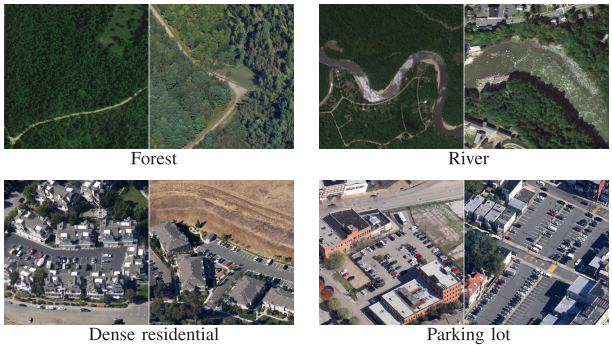


Fig. 4. Examples of major confusion in the RSSCN7 data set.

as dense residential/parking lot, and forest/river. As shown in Fig. 4, the forest and river share similar backgrounds. The dense residential and parking lots of scenes in Fig. 4 are composed of the same objects, such as roads and buildings, which can affect the feature representation of the scenes.

2) Results on AID→RSSCN7: Our second experiment was conducted on AID→RSSCN7. Table I shows the OA and Kappa gained by two different features. A similar tendency is observed: CS-DDA still outperforms over the state-of-the-art. Furthermore, from Table I, we can discover that the OA and Kappa of almost all methods improve compared to NN. This phenomenon illustrates that domain shift also appears in AID and RSSCN7 data sets.

In the first case of AlexNet as a feature extractor, for "Forest" and "River," CS-DDA achieves the best CA compared with the state-of-the-art methods. In Fig. 5, we provide the confusion matrix for both scenarios in AID→RSSCN7. The confusion on AID→RSSCN7 is similar to the confusion on NWPU→RSSCN7.

3) Results on UC Merced→RSSCN7: Our third experiment was conducted on the UC Merced→RSSCN7, and the results are provided in Table I. From Table I, it can be seen that CS-DDA achieves significantly better performance than the state-of-the-art methods.

In the first scenario of AlexNet as a feature extractor, for "Dense Residential" and "Farmland," CS-DDA realizes the best CA compared with the state-of-the-art methods. In Fig. 6, we provide the confusion matrix for both scenarios in UC

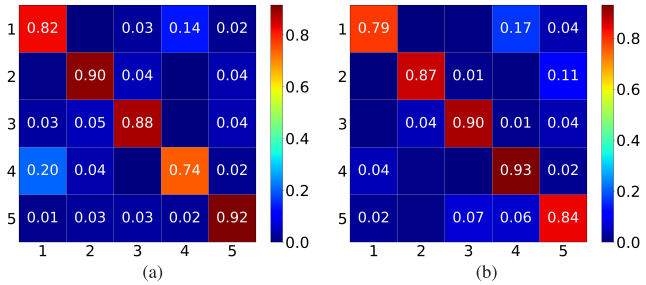


Fig. 5. Confusion matrix on the AID→RSSCN7. (a) AlexNet as a feature extractor. (b) ResNet50 as a feature extractor. The rows and columns of the matrix denote the actual and predicted classes, respectively. The corresponding category labels of 1–5 are dense residential, farmland, forest, parking lot, and river. The color bar indicates the proportion of samples over the actual total class samples.

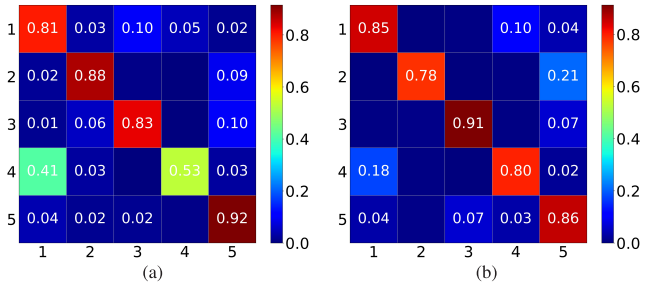


Fig. 6. Confusion matrix on the UC Merced→RSSCN7. (a) AlexNet as a feature extractor. (b) ResNet50 as a feature extractor. The rows and columns of the matrix denote the actual and predicted classes, respectively. The corresponding category labels of 1–5 are dense residential, farmland, forest, parking lot, and river. The color bar indicates the proportion of samples over the actual total class samples.

Merced →RSSCN7. Compared with the previous two groups of experiments, the result of UC Merced as the source domain is a little worse. The reason is that UC Merced is much more simple than AID and NWPU.

Generally speaking, the experiments on three cross-domain remote sensing image scene data sets show similar results. CS-DDA is designed to not only align the source and target domains as much as possible but also dynamically adjust the marginal distribution and conditional distribution.

D. Feature Distribution Analysis

In this section, to understand more deeply about CS-DDA, we provide the feature distributions of UCM→RSSCN7 in the 2-D space. We leverage the t-distributed stochastic neighbor embedding (t-SNE) [52] as a visualization tool to observe the data shift between the source domain and the target domain.

The feature distribution of Fig. 7 is original without DA. From Fig. 7, we can observe that the source domain and the target domain have large data shifts. The feature distribution of Fig. 8 is processed by CS-DDA. It is clear that CS-DDA alleviates the distribution discrepancy between the two domains. In Fig. 8, we can observe that CS-DDA is more discriminative, which implies that the source classifier will predict the target data more correctly. In addition, CS-DDA aligns the corresponding categories of the source

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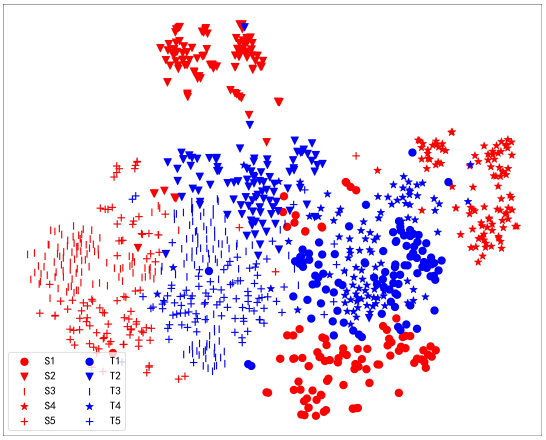


Fig. 7. Feature visualization: data embedding by t-SNE of original features. Red: source domain. Blue: target domain. Samples in different shapes represent that they are in different categories. The corresponding label categories in the legend are dense residential, farmland, forest, parking lot, and river.

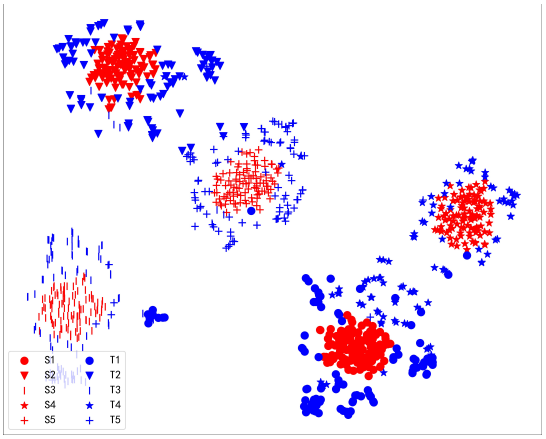


Fig. 8. Feature visualization: data embedding by t-SNE of CS-DDA features. Red: source domain. Blue: target domain. Samples in different shapes represent that they are in different categories. The corresponding label categories in the legend are dense residential, farmland, forest, parking lot, and river.

domain and the target, which is also beneficial to classify the target domain.

IV. Conclusion

In this article, we introduce a CS-DDA method to alleviate the data shift in remote sensing scene classification. The algorithm of CS-DDA includes SCM and DSDA. Among them, SCM generates a beneficial feature representation, and DSDA eliminates distribution discrepancies between domains. The major contributions of this article consist of three folds: 1) we present a new feature distribution alignment method based on DA; 2) to prevent mapping the source domain data into unrelated subspaces, we propose SCM; and 3) since the remote sensing image has the characteristics of intraclass diversity and interclass similarity, we leverage a balance factor to adjust the class condition distribution. We exploit the features of two classic pretrained CNN models to experiment on three cross-domain data sets. The experimental results demonstrate that the proposed method is superior to several state-of-the-art methods.

In future, we will extend the CS-DDA method to confront different DA scenarios in remote sensing scene classification.

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Jun Zhang received the B.S. and Ph.D. degrees from the Hebei University of Technology, Tianjin, China, in 1999 and 2011, respectively.

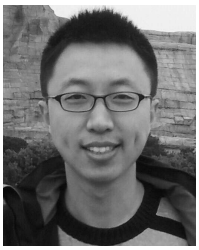
He is an Associate Professor with the School of Artificial Intelligence, Hebei University of Technology. His research interests include machine learning and intelligent computing.



Jiao Liu received the B.S. degree in computer science and technology from Taishan Medical University, Tai'an, China, in 2018. She is pursuing the M.S. degree with the School of Artificial Intelligence, Hebei University of Technology, Tianjin, China.

Her research interests include machine learning and intelligent computing.

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Bin Pan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Astronautics, Beihang University, Beijing, China, in 2013 and 2019, respectively.

Since 2019, he has been an Associate Professor with the School of Statistics and Data Science, Nankai University, Tianjin, China. His research interests include machine learning, remote sensing image processing, and multiobjective optimization.



Zhenwei Shi (Member, IEEE) received the Ph.D. degree in mathematics from the Dalian University of Technology, Dalian, China, in 2005.

From 2005 to 2007, he was a Post-Doctoral Researcher with the Department of Automation, Tsinghua University, Beijing, China. From 2013 to 2014, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, Northwestern University, Evanston, IL, USA. He is a Professor and the Dean of Image Processing Center, School of Astronautics, Beihang University, Beijing. He has authored or coauthored more than 100 scientific articles in related journals and proceedings, including the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, the IEEE TRANSACTIONS ON NEURAL NETWORKS, the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the IEEE Conference on Computer Vision and Pattern Recognition. His research interests include remote sensing image processing and analysis, computer vision, pattern recognition, and machine learning.

Dr. Shi received the Best Reviewer Awards for his service to the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2017. He has been an Associate Editor for the *Infrared Physics and Technology* since 2016.

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